

Renewable Energy — Module 2 Study Guide: Renewable Energy Options & Technological Assessment

Course Macro-Framing: Energy Transitions & Resource Dynamics

As human civilization expands, our global energy demand continues to accelerate. Transitioning from finite fossil fuels to renewable energy requires understanding both the physics of energy conversion and the economic mechanisms driving adoption:

FOSSIL COMMODITY PARADIGM			RENEWABLE INFRASTRUCTURE MODEL	
• Continuous daily fuel purchases	—>		• High upfront capital investment	
• Susceptible to fuel supply shocks	—>		• Free natural inputs (Sun, Wind, Water)	
• CO ₂ & atmospheric emissions	—>		• Zero operational air pollution	
• Resource depletion & price volatility			• Reaches grid parity over time	

1. **The Grid Parity Objective:** Renewable technology reaches economic maturity when it achieves **grid parity**—the threshold where generating renewable power becomes equal to or less expensive than buying electricity generated from traditional fossil fuels.
2. **Capital vs. Commodity:** While fossil fuels demand continuous commodity purchases, renewable systems require upfront capital investments (CapEx) to install infrastructure (solar PV arrays, hydraulic turbines, tidal barrages). Once deployed, the natural energy input is free and continuously replenished.
3. **Global Scale & Population Drivers:** World population is expanding from over 7 billion to over 8 billion people. Meeting global needs requires systematic **renewable energy mapping** to identify geographical energy assets and deploy utility-scale solutions.

Lesson 1: Solar Energy & Lesson 2: Mechanics of Solar Energy

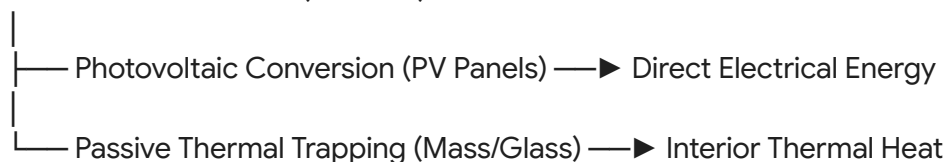
Level 1: The Basics (Foundational Concepts)

- [] **Mastery Checklist:** Define solar energy, grid parity, passive energy design, and identify global adoption leaders and key physical constraints.

What is Earth's most abundant energy source, and how do we capture it?

- **Solar Energy:** Radiant light and thermal energy derived directly from nuclear fusion inside the sun. It is officially classified as Earth's **most abundant and least used renewable energy source**.
- **Grid Parity:** The point at which the cost of generating electricity from a renewable energy source is **equal to or less expensive** than electricity generated by traditional fossil fuels.
- **Global Leadership: China** holds the world record for the most total solar energy installations globally.
- **Key Physical Constraint (Space):** A primary disadvantage of solar energy is the large amount of land or surface **space** required to deploy solar collector arrays. Solar power produces **no noise, no direct air pollution, and no operational greenhouse gases**.

SOLAR RADIANT ENERGY (Photons)



MIND-BLOWING FACT: The sun delivers more solar energy to Earth in a single hour than all of human civilization consumes in an entire year!

EXAM TRAP ALERT: If an exam question asks for a disadvantage of solar energy, do NOT select noise, air pollution, or greenhouse gases! Solar power generates zero operational noise or emissions. The correct disadvantage is **space** (land footprint).

EXAM CHEAT CODE:

- Cost equals or beats fossil fuels = **Grid parity**
- Nation with most solar installations = **China**
- | Most abundant & least used renewable = **Solar energy**
- Primary disadvantage of solar = **Space**

Level 2: Deepening Understanding (Conversion Mechanics & Design)

- **Photovoltaic Conversion:** Solar electric power is created when the **radiant energy** of sunlight strikes semiconductor material (like silicon), releasing electrons to produce **electrical energy**.
- **Passive Energy Design:** Architectural design that naturally traps the sun's thermal heat

within building mass (concrete floors, specialized glass) to warm interior spaces **without mechanical pumps or electric fans**.

- **Thermodynamics:** The branch of physical science that studies how energy is transferred and converted from disordered heat into usable, ordered forms (like electricity or work).

COVER & TEST YOURSELF:

1. What term describes the economic point where solar power becomes as cheap as coal or gas electricity?

[Answer: Grid parity.]

2. Which architectural strategy warms a building by trapping solar heat without using mechanical equipment?

[Answer: Passive energy design.]

Level 3: Expert Analysis (Socio-Economic Application)

Real-World Connection (Solar Microgrids in Rural Kenya): In off-grid communities across East Africa, solar photovoltaic systems paired with mobile payment technology allow households to bypass expensive fossil fuel grid extensions. By achieving local **grid parity** against diesel generators, localized solar power delivers immediate economic mobility.

Lesson 3: Hydro Energy & Lesson 4: The Mechanics of Hydro Energy

Level 1: The Basics (Foundational Concepts)

- [] **Mastery Checklist:** Trace waterpower history from ancient waterwheels to Benoît Fourneyron, define reservoirs and hydraulic head, and contrast hydro project designs.

How do we convert the natural flow of water into utility-scale grid power?

HYDROLOGICAL CYCLE (Evaporation & Rain) —► ELEVATED WATER RESERVOIR (Potential Energy)



ELECTRICITY TO GRID ◀— GENERATOR ◀— HYDRAULIC TURBINE ◀— FALLING WATER (Kinetic Energy)

Historical Milestones & Core Infrastructure

1. **Ancient Waterwheels:** Ancient civilizations first harvested waterpower using the **waterwheel** to grind grain and lift irrigation water.
2. **Benoît Fourneyron (1827):** French engineer who designed an outward-flow hydraulic turbine that was significantly more efficient than traditional waterwheels, laying the

groundwork for modern hydroelectric generators.

3. **Reservoir:** A large, man-made lake created behind a hydroelectric dam to store water for power generation and water management.
4. **Head:** The **vertical distance** water falls between the intake level (reservoir surface) and the turbine below.

MIND-BLOWING FACT: Hydroelectric facilities can go from completely shut down to full power output in just a few minutes, making them essential for balancing rapid changes in electrical grid demand!

EXAM TRAP ALERT: Do not confuse *head* with the channel or shaft! *Head* specifically means the **vertical distance** the water falls between the intake and the turbine.

EXAM CHEAT CODE:

- Waterpower first harvested using = **Waterwheel**
- Outward-flow efficient turbine designer = **Benoît Fourneyron**
- Large man-made lake = **Reservoir**
- Vertical distance water falls = **Head**

Level 2: Deepening Understanding (Facility Types & Impact Analysis)

Hydroelectric Facility Classifications

- **Diversion System (Run-of-River):** Hydroelectric design that channels a portion of a river's natural flow through a canal or penstock. It is the **least disruptive to the natural environment** because it does not require flooding large valleys.
- **Pumped Storage System:** Hydroelectric structure that uses **two reservoirs** at different elevations. Water is pumped to the upper reservoir during low demand and released back down through turbines during peak hours.
- **Socio-Environmental Impacts of Dams:** While hydroelectricity produces zero direct operational emissions, building large dams creates three major challenges: **relocation of people, landslides, and water pollution.**

HYDROELECTRIC SYSTEM COMPARISON MATRIX

System Type	Facility Structure	Environmental Impact	Key Distinction
Diversion System	River channel bypass	Lowest impact	No large dam or valley flooding required
Impoundment System	Dam & single reservoir	High impact	Floods land, creates large man-made reservoir
Pumped Storage	Two reservoirs	Moderate impact	Acts as a grid energy storage battery

COVER & TEST YOURSELF:

1. Which hydroelectric design is least disruptive to the natural environment?

[Answer: Diversion system.]

2. What three major socio-environmental problems occur when implementing large dams?

[Answer: Relocation of people, landslides, and water pollution.]

Level 3: Expert Analysis (Engineering Dynamics)

Hydroelectric power output is determined directly by two physical factors: flow rate and vertical head drop.

Hydroelectric Power is directly proportional to Flow Rate $\times$ Head (Vertical Drop).

By maximizing the **head** distance, engineers can generate massive power using smaller volumes of water, optimizing output across seasonal river fluctuations.

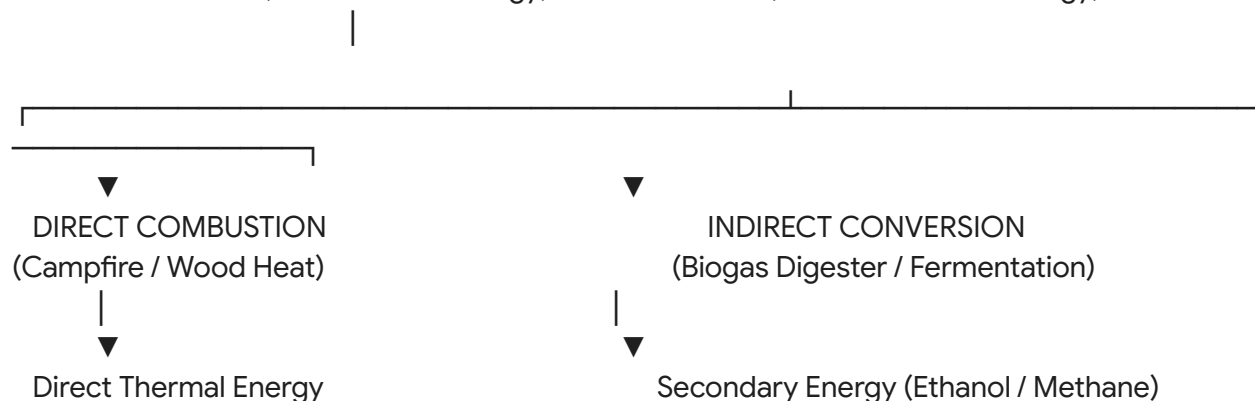
Lesson 5: Bioenergy & Lesson 6: The Mechanics of Bioenergy

Level 1: The Basics (Foundational Concepts)

- [] **Mastery Checklist:** Define bioenergy, biomass, biogas digesters, cogeneration, secondary energy vectors, and cite historical energy data.

How do we release stored chemical energy from organic materials?

PHOTOSYNTHESIS (Solar Radiant Energy) $\longrightarrow$ BIOMASS (Stored Chemical Energy)



Core Concepts for Exam Mastery:

1. **Campfire:** A direct, everyday example of bioenergy (combustion of wood biomass releasing heat and light).
2. **Pre-Industrial Biomass:** Before the Industrial Revolution, **wood** made up the majority of

biomass used worldwide for heating and cooking.

3. **Biogas Digesters: Airtight containers** used to ferment organic waste (livestock manure, sewage, crop residues) via anaerobic bacteria to produce methane gas.
4. **1876 Ethanol Engine:** In **1876**, the first internal combustion engine capable of running on ethanol-blended gasoline was developed.
5. **Cogenerated Energy:** Energy that is captured and **used by the same industry or facility that created it** (also known as Combined Heat and Power, CHP).
6. **Secondary Energy:** Energy that has been converted from raw biomass into a usable, refined fuel vector (such as liquid ethanol, biodiesel, or purified biomethane).
7. **U.S. Transportation Share:** Approximately **one-fourth** (25%) of the biomass supply in the United States is used by the transportation sector to produce ethanol and biodiesel.

MIND-BLOWING FACT: Biogas digesters capture methane that would otherwise escape into the atmosphere from decaying waste, turning a potent greenhouse gas into clean cooking and heating fuel!

EXAM TRAP ALERT: Remember exact figures and dates for the exam:

- First ethanol-blended combustion engine invented in = **1876**
- Share of U.S. biomass used for transportation biofuels = **One-fourth**
- Primary pre-industrial biomass resource = **Wood**

EXAM CHEAT CODE:

- Everyday bioenergy example = **Campfire**
- Airtight containers fermenting waste to methane = **Biogas digesters**
- Fuel converted from biomass into liquid/gas = **Secondary energy**
- Energy used by the industry that created it = **Cogenerated**

Level 2: Deepening Understanding (Biomass Conversion Pathways)

COVER & TEST YOURSELF:

1. What fraction of total U.S. biomass is used to create transportation biofuels like ethanol?
[Answer: One-fourth.]
2. What term describes waste heat or energy captured and used by the very facility that produced it?
[Answer: Cogenerated energy.]

Level 3: Expert Analysis (Multicultural Application)

Real-World Connection (Biogas Digesters in Rural Vietnam): In small-scale agricultural

farms across the Mekong Delta, livestock manure is directed into household **biogas digesters**. The airtight digesters produce methane gas for home cooking fuel, replacing harvested wood and reducing indoor air pollution while providing nutrient-rich organic fertilizer.

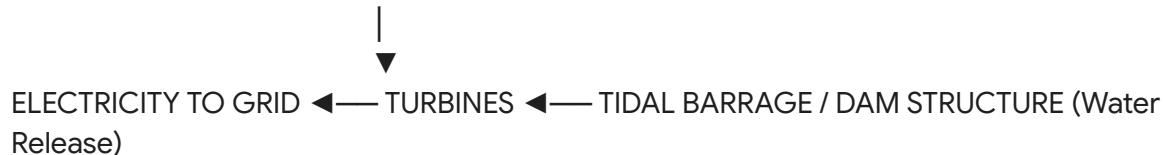
Lesson 7: Ocean Energy & Lesson 8: The Mechanics of Ocean Energy

Level 1: The Basics (Foundational Concepts)

- [] **Mastery Checklist:** Trace tidal mill history to 787 AD, define tidal siting rules (16-foot range), identify global megaprojects, and classify tidal energy states.

How do ocean tides turn lunar gravity into grid electricity?

LUNAR & SOLAR GRAVITATIONAL PULL —► OCEAN TIDE ELEVATION SHIFT (Kinetic & Potential Energy)



Historical Milestones & Siting Criteria:

1. **Earliest Tide Mills (787 AD):** The earliest tide mills were constructed on the coasts of Spain, France, and Great Britain as early as **787 AD** (early medieval era).
2. **World's Largest Tidal Plant: South Korea** operates the world's largest tidal power plant (the Sihwa Lake Tidal Power Station, producing 254 MW).
3. **Prior Energy State:** Before being converted into electricity, flowing tidal power exists as **kinetic energy** (and gravitational potential energy).
4. **The 16-Foot Minimum Rule:** Not all coastal areas are suitable for tidal power. To harness tidal energy economically, there must be a vertical difference of **at least 16 feet** (approximately 5 meters) between high tide and low tide.
5. **Tidal Conversion Structure:** The primary macro-structure built across an estuary or bay to hold back water and convert tidal energy into electricity is a **dam** (also called a *tidal barrage*).

MIND-BLOWING FACT: Ocean tides are predictable hundreds of years in advance because they are driven by the orbital mechanics of the Moon and Sun, making tidal energy one of the most reliable renewable energy sources on Earth!

EXAM TRAP ALERT: Pay attention to dates and numbers:

- Earliest coastal tide mills built in = **787** (787 AD)

- Minimum required difference between high and low tide = **16 feet**
- Structure used to convert tidal power = **Dams**
- Nation with the world's largest tidal power plant = **South Korea**

EXAM CHEAT CODE:

- Earliest tide mills date = **787**
- Country with largest tidal plant = **South Korea**
- Energy state of tides before electricity = **Kinetic energy**
- Minimum tide difference required = **16 feet**
- Primary structure used for tidal power = **Dams**

Level 2: Deepening Understanding (Tidal Barrage Mechanics)

COVER & TEST YOURSELF:

1. What is the minimum required vertical height difference between high and low tide to run a tidal plant economically?

[Answer: 16 feet.]

2. Which country operates the largest tidal power plant in the world?

[Answer: South Korea.]

Level 3: Expert Analysis (Marine Siting & Engineering)

Tidal power generation relies on water volume flowing through barrage channels. Building a tidal **dam** across an estuary allows engineers to control water flow during low tide, directing high-pressure water through submerged turbines to generate clean electricity.

Lesson 9: Measures of Success & Lesson 10: Technological Promise

Level 1: The Basics (Foundational Concepts)

- [] **Mastery Checklist:** Define renewable energy mapping and explain global demographic energy demand drivers.

How do strategic planners map clean energy assets to power our growing world?

- **Renewable Energy Mapping:** The systematic analytical process of identifying where renewable resources (solar, wind, hydro, geothermal, tidal) are geographically located and assessing their potential contribution to a nation's energy grid.
- **Global Demographic Projection:** Over the next twenty years, the world's population is

projected to expand from **over 7 billion people to over 8 billion people** (and toward 9 billion). This population expansion increases baseline global energy demand, driving the need for rapid renewable energy mapping and utility-scale deployment.

GEOSPATIAL & SATELLITE DATA —► RENEWABLE ENERGY MAPPING —► TARGETED CLEAN GRID DEPLOYMENT

EXAM CHEAT CODE:

- Process identifying resource locations and energy potential = **Renewable energy mapping**
- Population projection over next 20 years = **Over 7 billion people to over 8 billion people**

Comprehensive Module 2 Review & Exam Quick-Reference

This reference table brings together every key fact, statistic, historical name, date, and term required for the Module 2 Exam:

Concept / Question Target	Key Term / Exact Value	Specific Exam Context & Contextual Anchor
Fossil Fuel Price Match	Grid parity	Point where renewable electricity equals or beats fossil fuel costs.
Global Solar Installation Leader	China	Country with the most installed solar energy capacity globally.
Most Abundant & Least Used	Solar energy	Earth's most abundant yet historically underutilized energy source.
Disadvantage of Solar Energy	Space	Large surface area footprint requirement (does NOT produce air pollution).
Solar Conversion Output	Electrical energy	Radiant light converted into electrical current via solar cells.
Non-Mechanical Solar	Passive energy design	Building orientation and

Trapping		material choice trapping solar heat naturally.
Science of Heat Conversion	Thermodynamics	Science branch studying conversion of heat into usable ordered energy.
Ancient Waterpower Technology	Waterwheel	First device used by ancient societies to harvest water power.
Outward-Flow Turbine Inventor	Benoît Fourneyron	French engineer who created the efficient 1827 hydraulic turbine.
Least Disruptive Hydro System	Diversion system	Run-of-river hydro setup that channels water without large dams.
Man-Made Lakes Term	Reservoir	Large artificial body of water created behind a hydroelectric dam.
Vertical Water Drop Distance	Head	Distance water falls between the intake and the turbine blades.
Two-Reservoir Hydro System	Pumped storage	Facility using upper and lower reservoirs for grid energy storage.
Socio-Environmental Dam Problems	Relocation of people, landslides, water pollution	Three major negative impacts of building large hydroelectric dams.
Direct Everyday Bioenergy	Campfire	Everyday example of direct biomass combustion releasing heat.
Airtight Methane Fermenter	Biogas digesters	Airtight vessels used to ferment organic waste into methane gas.

Ethanol Blend Engine Date	1876	Invention year of the first engine capable of running on ethanol fuel.
Pre-Industrial Biomass Fuel	Wood	Primary biomass resource used worldwide before the Industrial Revolution.
Self-Used Industrial Energy	Cogenerated	Energy captured and used by the same facility that produced it.
Converted Biofuel State	Secondary energy	Biomass converted into usable liquid or gaseous fuel vectors.
U.S. Biofuel Transportation Share	One-fourth	25% share of U.S. biomass used for ethanol and biodiesel production.
Earliest Tide Mills Date	787	Earliest recorded tide mills built on European coasts (787 AD).
Largest Tidal Plant Nation	South Korea	Country operating the world's largest tidal plant (Sihwa Lake).
Tidal Energy Prior State	Kinetic energy	Motion state of tidal currents prior to electricity generation.
Minimum Tidal Range Rule	16 feet	Minimum vertical difference between high/low tide for tidal energy.
Tidal Conversion Structure	Dams	Structural barrages built across estuaries to convert tidal power.
Resource Location Process	Renewable energy mapping	Process mapping resource availability and energy potential.

Global Population Shift	Over 7 billion people to over 8 billion people	Demographic population projection driving global clean energy demand.
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If you want to know more about this, visit these sites

Explores interactive solar cell physics, semiconductor photon absorption, and photovoltaic power conversion.

- [PhET Interactive Simulations — Solar Photovoltaics](#)

Provides authoritative water resource information, hydroelectric turbine mechanics, and streamflow data.

- [U.S. Geological Survey \(USGS\) — Water Science School](#)

Offers technical resources and research updates covering bioenergy, anaerobic digestion, and biofuels.

- [U.S. Department of Energy \(DOE\) — Bioenergy Technologies Office](#)

Maintains international data, research papers, and technology maps for wave, tidal, and ocean thermal energy.

- [Ocean Energy Systems \(OES\) — International Energy Agency](#)